Assessment and Capstone Guide

AI and Data Science Professional | What you hand in, when, and how the work is judged | Updated 4 November 2026

Status of this guide
Some rules below are proposed and are marked "To be decided by the tutor". Everything else describes what the sessions and the notebooks do today.
An earlier draft carried percentage weights. They had no source and are not used here.

1. What is assessed, and when

Part	When	What	How it counts
Exit check	End of every session	Three questions and short answers on the last slides.	Not marked. It shows what to revisit.
Check cells	In every lab	Cells that print OK or NOT YET (section 2).	Not marked. They show that your lab worked.
Checkpoint 1	Session 8	Cleaned notebook, data cleaning log, reconciliation note and files.	To be decided by the tutor. Marking is not yet fixed.
Checkpoint 2	Session 13	A one-page analysis report.	To be decided by the tutor.
Checkpoint 3	Session 19	A tested model beside a baseline model.	To be decided by the tutor.
Capstone	Question in 13, dataset in 14, model in 19, app in 23, talk in 24	A question, cleaned data, a model with a baseline, limits, a small app with a model card, and a ten-minute talk.	Judged with the rubric in section 5. How it counts is to be decided by the tutor.
Participation	All sessions	Labs run to the last task. Attendance.	To be decided by the tutor.

To be decided by the tutor
How Checkpoint 1 is marked: pass or fail, a score, or a rubric.
Whether the checkpoints and the capstone carry weights, what the weights are, and the pass mark for the certificate.
How attendance and lab completion count.
The dates by which each hand-in is due.
Until these are set, treat the hand-ins in this guide as the evidence the tutor will look at, and ask your tutor how they will count.

2. What the check cells mean and do not mean

Every Python notebook has check cells. After each answer you run a line such as check("label", value, expected). It prints OK or NOT YET, and the last cell, show_checks(), prints "All N checks pass" or "k of N checks pass".

What OK means	Your value is within a small tolerance of the expected value (0.01 unless the lab says otherwise). Your cell ran and produced that number.
What NOT YET means	Your value is outside the tolerance, or the cell has not been filled in, or it raised an error. It means look again. It is not a failure grade.
What a check cannot tell you	That you understand the method, that your reasoning is right, or that you did not copy the answer. A correct number can come from a wrong method, and a wrong number can come from a harmless rounding or a setup problem.
Only the checks you ran count	"All 7 checks pass" means the 7 checks you ran passed. A check you skipped is not counted. Run the cells in order and look at the total.
Libraries differ a little	Models from scikit-learn can differ from the slides by a few records or a unit in the second decimal. The checks use a tolerance for this. Random exercises (Session 4) use ranges.

A learner notebook starts with NOT YET	That is expected. The answer cells hold answer = None until you fill them in. The solution notebook ends with "All N checks pass".
Sheets sessions	Sessions 1 to 4 and 24 have a "Check your numbers" table on the slides. You compare your sheet with it by eye. There are no check cells.

3. The three checkpoints

Checkpoint 1, Session 8: clean a raw file and reconcile it

A raw order export is cleaned one defect at a time. Hand in four things.

You hand in	It passes when
The cleaned notebook	Every check cell says OK and the run ends with All checks pass.
The data cleaning log	It has one row for each of the seven steps, and the last row ends at 41,167.
A reconciliation note	One or two sentences name the reconciliation difference (132,690 UGX) and its likely cause.
The files orders_clean.csv and cleaning_log.csv	They are saved from df and the log, and open in Excel or Sheets.

A reconciliation difference you can explain is worth more than one you have hidden. The explanation in your note must be in your own words. How Checkpoint 1 is marked: to be decided by the tutor.

Checkpoint 2, Session 13: a one-page analysis report

One page. A manager who has never seen the data should understand it. You also write your capstone question in this session (section 4).

Section	What you write
Question	One sentence naming the target, the predictors and the decision.
Model	The multiple regression and the unit of every predictor.
Findings	Three coefficients as sentences, with intervals, and R squared.
Check	What the residual plot showed and what you would do about it.
Limits	What the model cannot prove, and the test RMSE against the baseline.

Checkpoint 3, Session 19: a tested model

Forecast clinic cases by district and month, then hand in your notebook and the results file it writes. It shows:

- a baseline (a seasonal naive forecast) and a model scored on months it never saw, split by date at 2025;
- the error in the units of the business (cases), reported with MAE, RMSE and MAPE;
- a scikit-learn Pipeline that packs the modelling job into one object;
- three sentences for a district manager: whether the model beats the baseline, by how much, and what could mislead.

The same three moves apply to your capstone: a baseline, a score on records the model never saw, and an error in business units. Show that you checked for data leakage.

4. The capstone

You choose a question and a dataset and carry them through the second half of the course. A good question names the target, the predictors, the data and the decision. Session 13 gives examples.

Session	Capstone step
13	Write and scope the question (Checkpoint 2)
14	Choose the dataset
19	Baseline and tested model (Checkpoint 3)

Session	Capstone step
23	Learn to publish the app and model card; apply it to your capstone before Session 24
24	Present for ten minutes

Dataset options

- Lakeside Mart: orders, order lines, branches, customers and weekly branch figures (a fictional retailer).
- Nile Microfinance loans: 8,000 loans, and whether each defaulted (fictional).
- District Clinics: monthly clinic visits by district (fictional).
- Customer feedback messages: 800 short messages (fictional).
- Your own question about one of the tables above. If you want to use data from your workplace, ask your employer in writing, remove names and phone numbers first, and never put it in a public repository.

What you hand in

1. A one-sentence question and the decision it serves.
2. The data cleaned, with a data cleaning log.
3. One analysis that answers part of the question.
4. One model with a baseline model and a metric chosen for the cost of a mistake, scored on records it has not seen.
5. The limits: what the data cannot show, who could be harmed by an error, and which groups you compared.
6. A small app (Session 23) with a model card and a data protection checklist.
7. A ten-minute presentation (Session 24).

5. The presentation and the rubric

Five slides in ten minutes: question (1 min), data (2 min), evidence (3 min), risk (2 min), decision (2 min). Evidence gets the most time. In the Session 24 lab your partner scores you from 1 to 3 on each row of the rubric below and writes one comment per row.

Criterion	Strong work (3)	Partly (2)	Not yet (1)
Question and evidence	A one-sentence question. Every claim points to data.	A question, but some claims lack data.	No clear question, or claims without data.
Metric and baseline	A metric chosen for the cost, on unseen data, beside a baseline model.	A metric, but no baseline.	One accuracy number, no baseline.
Limits and fairness	Limits first. Risks named. Groups compared. Data duties checked.	Some limits, said late.	Limits missing or hidden.
Clarity	Five slides, ten minutes, plain words, one decision.	Mostly clear, a little over time.	Over time, jargon, no decision.

A rubric is a marking guide: the same rows, with a description of each level, applied to every piece of work. Four rows with a top score of 3 give a total out of 12. Whether the partner score, the tutor score, or both count towards the certificate is to be decided by the tutor.

6. Working rules

- Use the course data. It is fictional and generated.
- A free chat assistant may explain an error or suggest code. Keep a short audit log: what you asked, what it gave, how you checked it, and whether you kept it. You must be able to explain every line you hand in.
- List what you did not check. A stated limit loses nothing. A hidden one does.
- Your published work is public. Check it for personal data, passwords and keys before you push it to GitHub.
- Never paste real customer or patient data into a notebook, an assistant or an app you share.